SANJAY JAYAKUMAR

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PROFILE SUMMARY

Data Engineer with over 1 year of hands-on experience in ETL pipeline development and GenAI/RAG systems using Apache Spark for structured and unstructured data processing. Proficient in PySpark, Databricks, SparkSQL, Airflow, and AWS/Azure fundamentals, with hands-on experience delivering reliable, analytics-ready datasets for reporting and forecasting. Contributed to data-driven projects for leading energy sector clients including ONGC, GAIL, NRL, and IGGL, supporting operational insights and workflow automation.

SKILLS

- **Programming:** Python, SQL
- **Data Engineering:** PySpark, SparkSQL, Databricks, Airflow, ETL
- **Cloud:** AWS (S3), Azure (Fundamentals)
- **GenAI:** Embeddings, FAISS, ChromaDB, LangChain, RAG
- **APIs & DevOps:** FastAPI, Docker, CI/CD

WORK EXPERIENCE

Planning Engineer — Data Engineering

Corrtech International Ltd., Guwahati, Assam. Apr 2025 – Present

- Developed and maintained scalable data solutions to support project planning, progress tracking, and operational reporting.
- Implemented data quality & consistency checks across structured and unstructured datasets.
- Optimized time-series analytics pipelines, reducing operational wastage by ~15%.
- Delivered analytics-ready datasets for downstream BI and forecasting systems.

Data Engineer Intern

Tringapps Research Labs, Chennai, Tamil Nadu. Jan 2024 – Mar 2024

- Developed LangChain-based autonomous agents for analytics task planning and execution.
- Automated analytics reporting workflows, improving team productivity by 25%.
- Supported data validation and anomaly detection across datasets.

PROJECTS

Autonomous Multi-Agent Executor

- Built an autonomous multi-agent system for retrieving and processing data from PDFs, CSVs, and APIs.
- Designed a modular architecture: ingestion → embeddings → vector store → semantic retrieval → LLM-based response generation using FAISS and ChromaDB.
- Improved contextual accuracy through optimized retrieval strategies and advanced prompt engineering.
- Deployed as a scalable Dockerized FastAPI service with CI/CD integration for real-time autonomous knowledge execution.

Multi-Source RAG QA System

- Developed a multi-source RAG pipeline over structured and unstructured data sources using FAISS and ChromaDB.
- Architecture: Ingestion → Embeddings → Vector Store → Retriever → LLM
- Improved answer accuracy using optimized retrieval and prompts.
- Implemented containerization, logging, and CI/CD pipelines to improve deployment reliability.

CERTIFICATION

Develop a RAG-based solution with your own data using Microsoft Foundry – Microsoft
Data Engineering on AWS Foundations – AWS
Amazon Simple Storage Service (Amazon S3) Performance Optimization – AWS

EDUCATION

B.E, Computer Science and Engineering 2020 – 2024

K. Ramakrishnan College of Technology, Tiruchirappalli